

Solution Requirements

Date	3 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Category	Requirement	Description
1	Authentication	No login required	The application is publicly accessible; no user account is needed to make predictions
2	Input Interface	Applicant detail form	System must accept 15 applicant inputs: gender, age, income, income type, employment years, occupation, education, family status, housing type, children count, family members, car ownership, property ownership, and contact flags
3	External interfaces	Kaggle dataset ingestion	Training pipeline must read application_record.csv and credit_record.csv from the data/ directory

4	Transactions processing	End-to-end prediction pipeline	System must preprocess inputs (encode + scale), run ML inference, and return result within 2 seconds
5	Reporting	Model comparison dashboard	Home page must display accuracy and ROC-AUC for all 4 trained models, highlighting the best model
6	Reporting	Prediction history	System must store and display the last 10 predictions made in the current browser session
7	Business Rules	Target variable logic	STATUS codes 1-5 (30+ days past due) → Rejected (0); codes 0, C, X → Approved (1)
8	Compliance	No PII storage	No personally identifiable information is stored; all prediction history is session-only and cleared on browser close

Non-Functional Requirements (NFR)

S.No	Category	Requirement	Target Metric
NFR-1	Performance	Prediction response time	< 2 seconds from form submission to result display

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NFR-2	Scalability	Concurrent users	Render auto-scales; application is stateless (session storage only)
NFR-3	Security	Data protection	No sensitive data stored on server; HTTPS enforced by Render
NFR-4	Reliability	Uptime	99% uptime on Render free tier; sleeps after 15 min inactivity
NFR-5	Usability	Responsive design	Mobile + desktop compatible via Bootstrap 5.3 responsive grid
NFR-6	Maintainability	Auto-deployment	Any push to GitHub master branch triggers automatic Render redeployment